

TCS-504: System Design — Assignment 1

4. Architectural Relationships & Lifetime Analysis

Entity Pair	Relationship Type	Design Justification
Cinema — Screen	Composition	Screens are created and owned by Cinema; cannot exist independently.
Screen — Seat	Composition	Physical seats belong exclusively to one physical screen.
Show — Movie	Aggregation	Show references a Movie; deleting a show does not remove the movie.
Show — Screen	Aggregation	Show borrows a Screen during a time window.
Show — ShowSeat	Composition	ShowSeat represents seat state for that specific show only.
Booking — Customer	Association	Customer and Booking exist independently.
Booking — ShowSeat	Aggregation	Booking references ShowSeats; cancelling booking releases them without deletion.
Booking — Payment	Composition	Payment is instantiated specifically for a single booking transaction.
Payment — Subclasses	Inheritance	UpiPayment, CardPayment, CashPayment inherit from abstract Payment.
BookingService — Booking	Composition	BookingService orchestrates, allocates, and owns all active bookings.